Meeting Summary: May 7th, 2025

Date: May 7th, 2025
Time: Not recorded in the meeting summary
Attendees: Prof. Jan Ciborowski and Feng Gu
Topic: Resolving contaminant data discrepancies and refining the use of extreme quantiles in ecological threshold analysis

Summary of Discussion

- Feng and Prof. Jan reviewed discrepancies between Feng's current chemical dataset and the earlier dataset used in prior work, with particular attention to the measurement units reported for PCBs, pesticides, and other contaminants.
- Prof. Jan explained that some contaminant variables should not be interpreted only as mass per sediment volume or weight because their ecological meaning depends on how they partition into organic matter. For oily contaminants such as PCBs and some pesticides, concentrations normalized by organic carbon can better reflect movement through the food web and biological exposure.
- They discussed why median regression is often insufficient for identifying ecological thresholds. Prof. Jan emphasized that upper quantiles are more informative when the goal is to detect limiting effects of contamination on ecological endpoints.
- The conversation also covered piecewise quantile regression, including the practical limitation that more extreme quantiles rely on fewer observations and therefore become less stable when sample size is limited.
- Feng noted that analyses based on the full dataset produced PCA loadings that differed from previous work. Prof. Jan suggested comparing the relevant data matrices cell by cell to determine whether the mismatch is caused by the underlying data or by differences in the analysis workflow.
- The meeting concluded with discussion of immediate priorities for Feng's conference presentation, website updates, literature review, and simulation work on the reliability of different quantile levels.

Decisions Made

- The chemical dataset will be rechecked carefully against the earlier dataset to resolve unit inconsistencies and identify any cell-level differences between Feng's version and the earlier data matrix.
- Future interpretation of sediment contamination will distinguish chemicals that should be normalized by organic carbon from those appropriately reported by general sediment mass or volume units.
- Feng's website and presentation materials will emphasize extreme quantiles rather than only median relationships when explaining ecological threshold detection.

- Sample size effects must be considered explicitly before selecting a very high quantile level for piecewise quantile regression.

Action Items

- **Feng:** Prepare summary slides for the conference presentation, starting with a concise overview slide to organize the talk.
- **Feng:** Update and finalize the website with corrected contaminant data, improved results, and revised homepage content highlighting extreme quantiles.
- **Feng:** Review the 2020 paper by Dr. Jabed Javed and Prof. Jan Ciborowski on defining ecological endpoints, with particular attention to the discussion of extreme quantiles.
- **Feng:** Conduct simulations to evaluate how sample size affects the reliability of threshold detection at different quantile levels, such as 80% and 90%.
- **Prof. Jan:** Review Feng's materials, annotate comments in blue and red, and send feedback by the weekend if possible.

Follow-Up

Tentative next meeting time: To be confirmed.

Supportive Materials

- The 2020 ecological endpoints paper by Dr. Jabed Javed and Prof. Jan Ciborowski was identified as key background reading for the quantile-threshold discussion.
- Feng's current website draft, conference presentation materials, and the contaminant datasets under comparison were the main working materials referenced in the discussion.